

# THE INTERNET OF THINGS



News, views and development on the Internet of Things

## “We think using open platforms that comply with standards is the right way to go...”

Internet of Things, Bangalore (IoTBLR) is a thriving do-it-yourself (DIY) community populated by enthusiasts working to create stuff for the Internet of Things (IoT). Nihal Kashinath, founder of the group and Nagasai Arun Panchakarla, co-organiser, tell us about the relevance of DIY for building the IoT

### JANANI GOPALAKRISHNAN VIKRAM

**How do you perceive your group as making a positive contribution towards the emerging IoT?**

**Nihal:** In the short term, creating awareness and building enthusiasm about the IoT amongst professionals will help drive the movement. As more people get involved, they learn from each other, come up with several great ideas, implement them and extend the IoT network.

In the long term, introducing kids to building things will make them excited about IoT and help make it sustainable. Along with all this, bringing in companies and commercialising technology will ensure that the economics work out, which is critical to the success of any large-scale movement. Our group is basically building the ecosystem to make all this happen.

**There are several boards available today for DIY projects, such as the Arduino and the Raspberry Pi, but I guess the components you require for IoT projects might have different requirements in terms of form factor, power consumption, amongst others? Can you provide some information on the kind of components you use?**

**Nagasai:** Yes the components required vary a lot amongst IoT projects,

especially to tackle the constraints of processing power, memory, power consumption, amongst others.

For initial prototyping, we are working on crowd-sourcing of the resources and boards such as Arduino, Raspberry Pi, Beagleboard, Telos B, amongst others.

For commercialised versions, we need to design our own cost-effective hardware solution based on MSP430, EFM32 and STM32 and low-power radio transceivers. Moore's Law will play a significant role in the IoT but not as it was portrayed, to improve processing power.

In the IoT case, it would be mostly used to improve power consumption and reducing cost of the hardware. So, to create an IoT product, we need to have custom hardware according to the requirements and design. And we need to create a knowledge base around that.

**There are many platforms today that claim to ease connectivity for IoT products—Contiki-based Thingsquare, Xively, amongst others. Do you think these are worth the bucks, or do you go with just plain vanilla operating systems for your projects?**

**Nihal:** I personally advocate building on others' work in this space, since they have already solved certain prob-

lems and you do not have to waste time and resources (funds, expertise and energy) solving the same problems again. Makes a huge difference when working with limited resources and when time-to-market is critical. For open source, that already happens fairly efficiently.

With proprietary/closed systems, there is an inherent loss of control over the project parameters. If that does not affect your design too much, go for it. Of course, it also depends on the resources you are willing to spend on this.

**Nagasai:** One needs to be aware of which licence they are using, and adhere to the open standards by IETF and IEEE, amongst others. As we look at history and the success of open platforms, I think using open platforms that comply with standards is the right way to go. Open platforms that allow commercialisation of the product will tremendously reduce the time required to bring the product to the market.

For example, Contiki operating system is a BSD-style licence, which allows commercialisation of the product. And the cloud platforms such as Thingsquare, Xively and Thingworx also provide APIs for data management, which bring the development cycle down from months to days.